

Problema 2: 50%

Para las siguientes CFGs realice lo solicitado, mostrando todo el procedimiento.

1.

$$S \rightarrow 0A0 \mid 1B1 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

2.

$$S \rightarrow aAa \mid bBb \mid \epsilon$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

3.

$$S \rightarrow ASA \mid aB$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b \mid \epsilon$$

- a) Elimine las producciones- ϵ .
- b) Elimine cualquier producción unaria de la gramática resultante.
- c) Elimine cualquier símbolo inútil de la gramática resultante.
 - a. Remueva símbolos que no producen.
 - b. Remueva símbolos no alcanzables.
- d) Coloque la CFG resultante en la Forma Normal de Chomsky (CNF).

1)
$$\begin{aligned} S &\rightarrow 0A0 \mid 1B1 \mid BB \\ A &\rightarrow C \\ B &\rightarrow S \mid A \\ C &\rightarrow S \mid \epsilon \end{aligned}$$

$\{S, A, B, C\}$

$$S_0 \rightarrow S \mid \epsilon$$

$$S \rightarrow 0A0 \mid 0S \mid 1B1 \mid 11 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S$$

Unitarias

Eliminar unitarias

$$S_0 \rightarrow S$$

$$S_0 \rightarrow \epsilon \mid 0A0 \mid 0C \mid 1B1 \mid 11 \mid BB$$

$S, C = \text{no alcanzables}$

$$S \rightarrow B$$

$$S \rightarrow 0A0 \mid 0C \mid 1B1 \mid 11 \mid BB$$

$$A \rightarrow C$$

$$A \rightarrow 0A0 \mid 0C \mid 1B \mid 1111BB$$

$\{S_0, A, B, C\}$ son productivas

$$B \rightarrow S$$

$$B \rightarrow 0A0 \mid 0C \mid 1B \mid 1111BB$$

$$B \rightarrow A$$

$$C \rightarrow 0A0 \mid 0C \mid 1B \mid 1111BB$$

$$C \rightarrow S$$

$$S_0 \rightarrow 0A0 \mid 0C \mid 1B1 \mid 11 \mid BB \mid \epsilon$$

$$A \rightarrow 0A0 \mid 0C \mid 1B1 \mid 11 \mid BB$$

$$B \rightarrow 0A0 \mid 0C \mid 1B1 \mid 11 \mid BB$$

$$X_0 \rightarrow 0$$

$$0A0 \rightarrow X_0 A X_0$$

$$0C \rightarrow X_0 C$$

$$X_1 \rightarrow 1$$

$$1B1 \rightarrow X_1 B X_1$$

$$11 \rightarrow X_1 X_1$$

$$Y_0 \rightarrow A X_0$$

$$0A0 \rightarrow X_0 Y_0$$

$$Y_1 \rightarrow B X_1$$

$$1B1 \rightarrow X_1 Y_1$$

CFG

$$S \rightarrow X_0 Y_0 \mid X_0 X_0 \mid X_1 Y_1 \mid X_1 X_1 \mid BB \mid \epsilon$$

$$A \rightarrow X_0 Y_0 \mid X_0 X_0 \mid X_1 Y_1 \mid X_1 X_1 \mid BB$$

$$B \rightarrow X_0 Y_0 \mid X_0 X_0 \mid X_1 Y_1 \mid X_1 X_1 \mid BB$$

$$Y_0 \rightarrow A X_0$$

$$Y_1 \rightarrow B X_1$$

$$X_0 \rightarrow 0$$

$$X_1 \rightarrow 1$$

2) $S \rightarrow aAa \mid bBb \mid \epsilon$
 $A \rightarrow C \mid a$
 $B \rightarrow C \mid b$
 $C \rightarrow CDE \mid \epsilon$
 $D \rightarrow A \mid B \mid ab$

$\{S, A, B, C, D\}$

Unarias

Eliminar unarias

$S_0 \rightarrow S \mid \epsilon$

$S_0 \rightarrow S$

$S_0 \rightarrow aAa \mid aabBb \mid bbb \mid \epsilon$

$E, C = \text{No productivos}$

$S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow C$

$S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow C \mid a$

$B \rightarrow C$

$A \rightarrow a \mid CDE \mid DE \mid CE \mid \epsilon$

$S_0, S, A, B, D = \text{Son productivos}$

$B \rightarrow C \mid b$

$C \rightarrow \epsilon$

$B \rightarrow b \mid CDE \mid DE \mid CE \mid \epsilon$

$C \rightarrow CDE \mid DE \mid CE \mid \epsilon$

$D \rightarrow A$

$C \rightarrow CDE \mid DE \mid CE$

$D \rightarrow A \mid B \mid ab$

$D \rightarrow B$

$D \rightarrow CDE \mid DE \mid CE \mid a \mid b \mid ab$

$S_0 \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$

$S_0 \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$

$S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow a$

$A \rightarrow a$

$B \rightarrow b$

$B \rightarrow b$

$D \rightarrow ab \mid a \mid b$

CFG

$S_0 \rightarrow AX \mid AA \mid BY \mid BB \mid \epsilon$

$X \rightarrow AA$

$Y \rightarrow BB$

$A \rightarrow a$

$B \rightarrow b$

3) $S \rightarrow ASA \mid aB$
 $A \rightarrow B \mid S$
 $B \rightarrow b \mid \epsilon$

Unarias

Eliminar unarias

$S \rightarrow ASA \mid SA \mid AS \mid aB \mid a$

$S \rightarrow S$

$S \rightarrow ASA \mid SA \mid AS \mid aB \mid a$

$S, A, B = \text{Son productivos}$

$A \rightarrow B \mid S$

$A \rightarrow B$

$A \rightarrow ASA \mid SA \mid AS \mid aB \mid a \mid b$

$B \rightarrow b$

$A \rightarrow S$

$B \rightarrow b$

CFG

$S \rightarrow AX_{SA} \mid SA \mid AS \mid X_a B \mid a$

$A \rightarrow AX_{SA} \mid SA \mid AS \mid X_a B \mid a \mid b$

$B \rightarrow b$

$X_{SA} \rightarrow SA$